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Complex Functions

INTRODUCTION The study of derivatives, integrals, and series can be extended from the real to the complex numbers, where it is called *complex analysis*. The study of complex functions is far from a straightforward extension of the study of real variable functions. Differentiable functions of a complex variable have many properties not found in their real relatives. For example, while we can find real functions that are once but not twice differentiable, this is not true for complex functions. If a complex function is differentiable once on an open region, then its second, third, and all higher order derivatives also exist. In Chapter 9 we saw differentiable functions that are not equal to any power series. In contrast, a complex differentiable function can always be expressed as a convergent power series. The study of complex functions gives profound new insights into the properties of the real-valued functions we have already encountered and opens the way to many new applications. The study of complex functions, particularly in higher dimensions, remains an active area of mathematical research. In this chapter we introduce the complex numbers and then study some of the many beautiful aspects of complex differentiable functions.

18.1 Complex Numbers

Complex numbers are expressed in the form $a + ib$, or $a + bi$, where a and b are real numbers and i is a symbol for a new type of number whose square equals -1 ; that is, $i^2 = -1$. We refer to a as the *real* part of the complex number and to b as the *imaginary* part. The words “real” and “imaginary” have connotations that tend to place $\sqrt{-1}$ in a less favorable position in our minds than a real number such as $\sqrt{2}$. In fact, a good deal of imagination, or inventiveness, was required to construct the real number system, which forms the basis of calculus (see Appendix A.7). We begin this chapter with a review of the successive stages of invented number systems.

The Hierarchy of Numbers

The first stage of number development was the recognition of the counting numbers $1, 2, 3, \dots$, which we now call the natural numbers or the positive integers. Certain arithmetical operations on the positive integers, such as addition and multiplication, keep us entirely within this system. That is, if m and n are any positive integers, then their sum $m + n$ and product mn are also positive integers. Some equations can be solved entirely within the system of positive integers. For example, we can solve $3 + x = 7$ using only positive integers. But other simple equations, such as $7 + x = 3$, cannot be solved if only positive integers are at our disposal. The invention of the number zero and the negative numbers allowed the solution of equations such as $7 + x = 3$. Using the integers

$$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$$

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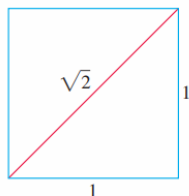


FIGURE 18.1 The diagonal of the unit square has irrational length.

we can always find the missing integer x that solves the equation $m + x = p$ when we are given the other two integers m and p in the equation. Addition and multiplication of integers always keep us within the system of integers. However, division does not, and so fractions m/n , where m and n are integers with n nonzero, were invented. The resulting system is called the rational numbers and is rich enough to perform all of the rational operations of arithmetic, including addition, subtraction, multiplication, and division (although division by zero is excluded since it is meaningless). Yet there are still simple polynomial equations that cannot be solved within the system of rational numbers. The ancient Greeks realized that there is no rational number that solves the equation $x^2 = 2$, even though the Pythagorean Theorem implies that the length x of the diagonal of the unit square satisfies this equation. (See Figure 18.1.)

To see why $x^2 = 2$ has no rational solution, consider the following argument. Suppose that there did exist some positive integers p and q with no common factor other than 1 such that the fraction $x = p/q$ satisfies $x^2 = 2$. Writing this out, we see that $p^2/q^2 = 2$, and therefore

$$p^2 = 2q^2.$$

Then p^2 is also an even integer. Since the square of an odd number is odd, we conclude that p itself must be an even number (for if p were odd, then p^2 would also be odd). So p is divisible by 2, and $p = 2k$ for some integer k . Hence $p^2 = 4k^2$. Since we already saw that $p^2 = 2q^2$, it follows that $2q^2 = p^2 = 4k^2$, and therefore $q^2 = 2k^2$.

Consequently, q^2 is an even number and this requires in turn that q itself be even. Therefore, both p and q are divisible by 2, which is contrary to our assumption that they contain no common factors other than 1. Since we have arrived at a contradiction, there cannot exist any such integers p and q , and therefore there is no rational number that solves the equation $x^2 = 2$.

The invention of real numbers addressed this issue (and others). Using real numbers, we can represent every possible physical length. Each real number can be represented as an infinite decimal $a.d_1d_2d_3d_4\dots$, where a is an integer followed by a sequence of decimal digits each between 0 and 9 (Appendix A.7). If the sequence stops or repeats in a periodic pattern, then the decimal represents a rational number. An irrational number is represented by a nonterminating and nonrepeating decimal. The rational and irrational numbers together make up the real number system. Unlike the rational numbers, the real numbers have the completeness property, meaning that there are no “holes” or “gaps” in the real line. This property is essential in calculus, implying for example that continuous functions on a closed interval have an absolute maximum value and an absolute minimum value (Extreme Value Theorem, Section 4.1), and that an increasing and bounded sequence converges (Monotonic Sequence Theorem, Section 9.1). Yet for all of its utility, there are simple equations that cannot be solved when working only within the real number system. As an example, the polynomial equation $x^2 + 1 = 0$ has no real solutions.

The Complex Numbers

We have discussed three systems of numbers that form a hierarchy, with each system containing the previous system. Each system is richer than its predecessor in that it permits additional operations to be performed without going outside the system.

1. Using the integer system, we can solve all equations of the form

$$x + a = 0, \quad (1)$$

where a is an integer.

2. Using the rational numbers, we can solve all equations of the form

$$ax + b = 0, \quad (2)$$

provided that a and b are rational numbers and $a \neq 0$.

3. Using the real numbers, we can solve all of Equations (1) and (2) and, in addition, all quadratic equations

$$ax^2 + bx + c = 0, \quad (3)$$

provided that $a \neq 0$ and $b^2 - 4ac \geq 0$.

The quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (4)$$

gives the solutions to Equation (3). When $b^2 - 4ac$ is negative, there are no real number solutions to the equation $ax^2 + bx + c = 0$. In particular, the simple quadratic equation $x^2 + 1 = 0$ cannot be solved using any of the three invented systems of numbers that we have discussed. Thus we are led to the fourth system of numbers, the set of complex numbers $a + ib$. The symbol i represents a new number whose square equals -1 . We call a the *real part* and b the *imaginary part* of the complex number $a + ib$.

Real and Imaginary parts

For a complex number $z = a + ib$, we write

$$\operatorname{Re} z = a$$

and

$$\operatorname{Im} z = b.$$

Sometimes it is convenient to write $a + bi$ instead of $a + ib$; both notations describe the same complex number. We define equality and addition for complex numbers in the following way.

Equality $a + ib = c + id$ if and only if $a = c$ and $b = d$.

Two complex numbers $a + ib$ and $c + id$ are equal if and only if their real parts are equal and their imaginary parts are equal.

Addition $(a + ib) + (c + id) = (a + c) + i(b + d)$

We sum the real parts and separately sum the imaginary parts.

To multiply two complex numbers, we multiply using the distributive rule and then simplify using $i^2 = -1$.

$$\begin{aligned} \text{Multiplication } (a + ib)(c + id) &= ac + iad + ibc + i^2bd \\ &= (ac - bd) + i(ad + bc) \quad i^2 = -1 \end{aligned}$$

EXAMPLE 1 Basic arithmetic operations with complex numbers

- a. The set of all complex numbers $a + i0$, where the second number b is zero, has all of the properties of the set of real numbers. For example, addition and multiplication as complex numbers give

$$(a + i0) + (c + i0) = (a + c) + i0, \quad (a + i0)(c + i0) = ac + i0,$$

which are numbers of the same type with imaginary part zero. We usually just write a instead of $a + i0$, and in this sense the complex number system contains the real number system.

- b. If we multiply a real number $a = a + i0$ by a complex number $c + id$, we get $a(c + id) = (a + i0)(c + id) = ac + iad$.

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- c. The number $0 = 0 + i0$ plays the role of zero in the complex number system, and the complex number $1 = 1 + i0$ plays the role of unity, or one.
- d. The complex number $i = 0 + i1$, which has real part zero and imaginary part one, has the property that its square is $i^2 = (0 + i1)^2 = (0 + i1)(0 + i1) = (-1) + i0 = -1$. So $x = i$ is a solution to the quadratic equation $x^2 + 1 = 0$. In the complex number system there are exactly two solutions to this equation, the other solution being $x = -i = 0 + i(-1)$.
- e. We can divide any two complex numbers, as long as we do not divide by the number $0 = 0 + i0$. When $a + ib \neq 0$ (meaning that either $a \neq 0$ or $b \neq 0$ or both $a \neq 0$ and $b \neq 0$), we carry out division as follows:

$$\frac{c + id}{a + ib} = \frac{(c + id)(a - ib)}{(a + ib)(a - ib)} = \frac{(ac + bd) + i(ad - bc)}{a^2 + b^2} = \frac{ac + bd}{a^2 + b^2} + i\frac{ad - bc}{a^2 + b^2}.$$

In this way we express the quotient in the standard form $x + iy$, where $x = (ac + bd)/(a^2 + b^2)$ and $y = (ad - bc)/(a^2 + b^2)$ are real numbers. Note that $a^2 + b^2 \neq 0$ since we stipulated that a and b cannot both be zero.

- f. The number $a - ib$ that was used in part (e) as the multiplier to clear the i from the denominator is called the *complex conjugate* of $a + ib$. If we denote the original complex number by $z = a + ib$, then it is customary to write $\bar{z}$ (read “z bar”) to denote its complex conjugate:

$$z = a + ib, \quad \bar{z} = a - ib.$$

Multiplying the numerator and denominator of a fraction $(c + id)/(a + ib)$ by the complex conjugate of the denominator will always replace the denominator by a real number. ■

EXAMPLE 2 Computations with complex numbers

- a. $(2 + 3i) + (6 - 2i) = (2 + 6) + (3 - 2)i = 8 + i$
- b. $(2 + 3i) - (6 - 2i) = (2 - 6) + (3 - (-2))i = -4 + 5i$
- c. $(2 + 3i)(6 - 2i) = (2)(6) + (2)(-2i) + (3i)(6) + (3i)(-2i)$
 $= 12 - 4i + 18i - 6i^2 = 12 + 14i + 6 = 18 + 14i$
- d. $\frac{2 + 3i}{6 - 2i} = \frac{(2 + 3i)(6 + 2i)}{(6 - 2i)(6 + 2i)} = \frac{12 + 4i + 18i + 6i^2}{36 + 12i - 12i - 4i^2} = \frac{6 + 22i}{40} = \frac{3}{20} + \frac{11}{20}i$ ■

The Complex Plane

We can use the usual xy -coordinate system to represent complex numbers as points or vectors in the plane. In this context we call it the *complex plane*. There are now two geometric representations of a complex number $z = x + iy$:

1. as a point $P(x, y)$ in the xy -plane,
2. as a vector $\overrightarrow{OP}$ from the origin to P .

In each representation, the x -axis is called the *real axis* and the y -axis is the *imaginary axis*, as in Figure 18.2. Geometric representations of complex numbers are sometimes called *Argand Diagrams*.

In polar coordinates we have

$$x = r \cos \theta, \quad y = r \sin \theta,$$

and

$$z = x + iy = r(\cos \theta + i \sin \theta). \quad (5)$$

The number 0, or $0 + i0$ is a special case, with $r = 0$ but no specified θ .

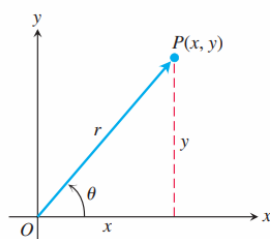


FIGURE 18.2 The complex number $z = x + iy$ can be represented both as a point $P(x, y)$ and as a vector.

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We define the *absolute value* (or *magnitude*) of a complex number $x + iy$ to be the length r of a vector $\overline{OP}$ from the origin to $P(x, y)$. We denote the absolute value by vertical bars; thus,

$$|x + iy| = \sqrt{x^2 + y^2}.$$

We always choose the polar coordinates r and θ so that r is nonnegative, and then

$$r = |x + iy|.$$

The polar angle θ is called the *argument* of z and is written $\theta = \arg z$. There is more than one choice for the argument. Any integer multiple of 2π may be added to θ to produce another appropriate angle. The following equation gives a useful formula connecting a complex number z , its conjugate $\bar{z}$, and its absolute value $|z|$:

$$z \cdot \bar{z} = |z|^2.$$

Euler's Formula

The identity

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (6)$$

is called Euler's Formula. The origins of this formula will be seen in Section 18.6. We show a vector representing $e^{i\theta}$ in Figure 18.3. Using Equation (6), we can write Equation (5) as

$$z = re^{i\theta}.$$

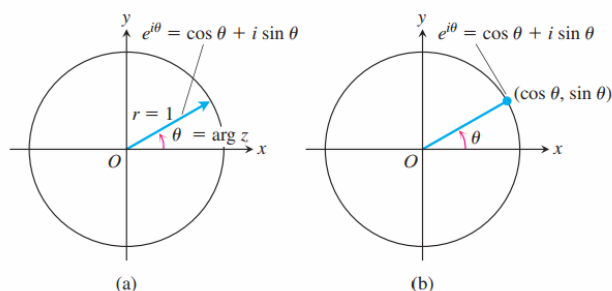


FIGURE 18.3 The complex number $z = e^{i\theta}$ as a vector and as a point.

The notation $\exp(A)$ is also used for e^A , so we could write $z = r \exp(i\theta)$.

Euler's formula leads to the following rules for calculating products, quotients, powers, and roots of complex numbers in polar form.

Products

To multiply two complex numbers in polar form, we multiply their absolute values and add their arguments. To see why, let

$$z_1 = r_1 e^{i\theta_1} \quad \text{and} \quad z_2 = r_2 e^{i\theta_2}, \quad (7)$$

so that

$$|z_1| = r_1, \quad \arg z_1 = \theta_1; \quad |z_2| = r_2, \quad \arg z_2 = \theta_2.$$

In the Cartesian $a + bi$ form we have

$$z_1 = r_1 \cos \theta_1 + ir_1 \sin \theta_1$$

and

$$z_2 = r_2 \cos \theta_2 + ir_2 \sin \theta_2.$$

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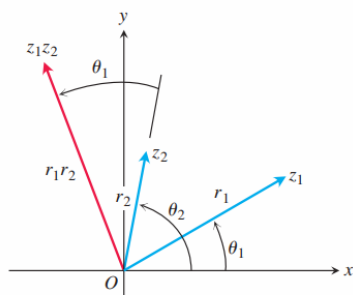


FIGURE 18.4 When z_1 and z_2 are multiplied, $|z_1 z_2| = r_1 \cdot r_2$ and $\arg(z_1 z_2) = \theta_1 + \theta_2$.

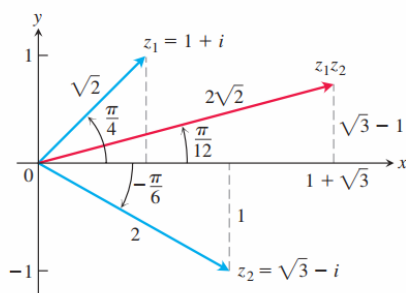


FIGURE 18.5 To multiply two complex numbers, multiply their absolute values and add their arguments.

Their product is then

$$\begin{aligned} z_1 z_2 &= r_1 e^{i\theta_1} r_2 e^{i\theta_2} \\ &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 ([\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2] + i [\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2]) \\ &= r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)) \quad \text{Angle sum formulas} \\ &= r_1 r_2 e^{i(\theta_1 + \theta_2)}. \end{aligned}$$

Therefore

$$r_1 e^{i\theta_1} r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}. \quad (8)$$

The absolute value of the product is $r_1 r_2$ and the argument of the product is the sum of the arguments of z_1 and z_2 . We have shown that the product of two complex numbers is represented by a vector whose length is the product of the lengths of the two factors and whose angle with the x -axis is the sum of the corresponding angles of the factors. (Figure 18.4.) In particular, Equation (8) implies that multiplying a vector by $e^{i\theta}$ causes its vector representation to be rotated counterclockwise through an angle θ . Multiplication by i rotates 90° , by -1 rotates 180° , by $-i$ rotates 270° , and so on.

EXAMPLE 3 We compute the product of $z_1 = 1 + i$ and $z_2 = \sqrt{3} - i$. We first plot these complex numbers in Figure 18.5, and we can then read off their polar representations

$$\begin{aligned} z_1 \cdot z_2 &= \sqrt{2} \exp\left(\frac{i\pi}{4}\right) \cdot 2 \exp\left(-\frac{i\pi}{6}\right) \\ &= 2\sqrt{2} \exp\left(\frac{i\pi}{4} - \frac{i\pi}{6}\right) \\ &= 2\sqrt{2} \exp\left(\frac{i\pi}{12}\right) \\ &= 2\sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12}\right) \\ &\approx 2.73 + 0.73i. \end{aligned}$$

EXAMPLE 4 **Computing Reciprocals** Show that if $z = re^{i\theta}$ and $r > 0$, then $\frac{1}{z} = \frac{1}{r} e^{-i\theta}$.

Solution Applying the formula for products, Equation (8), we see that

$$(re^{i\theta})\left(\frac{1}{r}e^{-i\theta}\right) = 1 \cdot e^{i0} = 1.$$

Dividing both sides by $z = re^{i\theta}$ gives

$$\frac{1}{r} e^{-i\theta} = \frac{1}{re^{i\theta}} = \frac{1}{z}.$$

Quotients

We now look at how to divide $z_1 = r_1 e^{i\theta_1}$ by $z_2 = r_2 e^{i\theta_2}$ in polar form. We assume that $r_2 > 0$ so that we are not dividing by zero. To compute the quotient z_1/z_2 we multiply z_1 by $1/z_2$. This gives

$$\frac{z_1}{z_2} = z_1 \left(\frac{1}{z_2}\right) = r_1 e^{i\theta_1} \left(\frac{1}{r_2} e^{-i\theta_2}\right) = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}, \quad \text{Equation (8)}$$



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so that

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

and

$$\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2 = \arg z_1 - \arg z_2.$$

That is, we divide lengths and subtract angles to compute the quotient of complex numbers.

EXAMPLE 5 Let $z_1 = 1 + i$ and $z_2 = \sqrt{3} - i$, as in Example 3. Then

$$\frac{1+i}{\sqrt{3}-i} = \frac{\sqrt{2}e^{i\pi/4}}{2e^{-i\pi/6}} = \frac{\sqrt{2}}{2}e^{5\pi i/12} \approx 0.707\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right) \approx 0.183 + 0.683i. \quad \blacksquare$$

Powers

If n is a positive integer, we can apply the product formulas in Equation (8) to find the polar form of a power

$$z^n = \underbrace{z \cdot z \cdot \cdots \cdot z}_{n \text{ factors}}$$

With $z = re^{i\theta}$, we obtain

$$\begin{aligned} z^n &= (re^{i\theta})^n = r^n e^{i(\theta + \cdots + \theta)} \\ &= r^n e^{in\theta}. \end{aligned} \quad \text{\textit{n summands}} \quad (9)$$

The length $r = |z|$ is raised to the n th power, and the angle $\theta = \arg z$ is multiplied by n . If we take $r = 1$ in Equation (9), we obtain De Moivre's Theorem.

De Moivre's Theorem

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad (10)$$

If we expand the left side of the equation in De Moivre's Theorem using the Binomial Theorem and reduce it to the form $a + ib$, we obtain trigonometric identities, namely formulas for $\cos n\theta$ and $\sin n\theta$ as polynomials of degree n in $\cos \theta$ and $\sin \theta$.

EXAMPLE 6 If $n = 3$ in Equation (10), we have

$$(\cos \theta + i \sin \theta)^3 = \cos(3\theta) + i \sin(3\theta).$$

The left side of this equation expands to

$$\cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta.$$

The real part of this must equal $\cos(3\theta)$ and the imaginary part must equal $\sin(3\theta)$. Therefore,

$$\begin{aligned} \cos(3\theta) &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta, \\ \sin(3\theta) &= 3 \cos^2 \theta \sin \theta - \sin^3 \theta. \end{aligned} \quad \blacksquare$$

Roots

If x is a positive real number, then it has exactly one real n th root if n is an odd integer and exactly two n th roots if n is even. If x is a negative real number, then x has exactly one real n th root if n is an odd integer and no real n th root if n is even. In contrast, if $z = re^{i\theta}$ is a



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complex number different from zero, then there are precisely n different complex numbers, $w_0, w_1, \dots, w_{n-1}$, that are n th roots of z . To see why, let $w = \rho e^{i\alpha}$ be an n th root of $z = re^{i\theta}$. Then

$$w^n = z$$

or

$$\rho^n e^{in\alpha} = re^{i\theta}.$$

Since both r and ρ are positive real numbers, this implies that $\rho^n = r$, and so

$$\rho = \sqrt[n]{r}$$

is the real, positive n th root of r . Although we cannot say that $n\alpha$ and θ must be equal, since the argument of a complex number is only defined up to multiples of 2π , we can say that they may differ only by an integer multiple of 2π . That is,

$$n\alpha = \theta + 2k\pi, \quad k = 0, \pm 1, \pm 2, \dots$$

Therefore,

$$\alpha = \frac{\theta}{n} + k \frac{2\pi}{n}.$$

This gives the following collection of n distinct complex numbers,

$$\sqrt[n]{re^{i\theta}} = \sqrt[n]{r} \exp\left[i\left(\frac{\theta}{n} + k \frac{2\pi}{n}\right)\right], \quad k = 0, \pm 1, \pm 2, \dots \quad (11)$$

Each of these has the property that its n th power equals z , so these n complex numbers give all of the n th roots of $z = re^{i\theta}$.

The expression $\sqrt[n]{re^{i\theta}}$ on the left of Equation (11) defines a new concept, the n th root of a complex number, while the term $\sqrt[n]{r}$ on the right refers to the real n th root. There is exactly one positive real number whose n th power equals the positive real number r , and that number gives the value of $\sqrt[n]{r}$. On the other hand, the expression $\sqrt[n]{re^{i\theta}}$ refers to a collection of n distinct numbers. In practice we often want to pick out a particular n th root, the *principal* n th root. We can do that by requiring that $z = re^{i\theta}$ have $-\pi < \theta \leq \pi$, and then taking the *principal* n th root to be

$$\sqrt[n]{r} \exp\left[i\left(\frac{\theta}{n}\right)\right].$$

There might appear to be infinitely many different answers in Equation (11), corresponding to the infinitely many possible values of k , but $k = m + n$ gives the same complex number as $k = m$. This is true because the complex number

$$\exp\left[i\left(\frac{\theta}{n} + 2\pi \frac{m+n}{n}\right)\right] = \exp\left[i\left(\frac{\theta}{n} + 2\pi \frac{m}{n} + 2\pi\right)\right]$$

and the complex number

$$\exp\left[i\left(\frac{\theta}{n} + 2\pi \frac{m}{n}\right)\right],$$

are equal, since they have arguments that differ by 2π . Therefore we need only take n consecutive values for k to obtain all the different complex numbers that are n th roots of z . For convenience, we take

$$k = 0, 1, 2, \dots, n-1.$$

All the n th roots of $re^{i\theta}$ lie on a circle centered at the origin and having radius equal to the real, positive n th root of r . One of them has argument $w_0 = \theta/n$. The others are uniformly spaced around the circle, each being separated from its neighbors by an angle equal to $2\pi/n$. (Figure 18.6 illustrates the placement of the three cube roots, w_0, w_1, w_2 , of the complex number $z = re^{i\theta}$.)

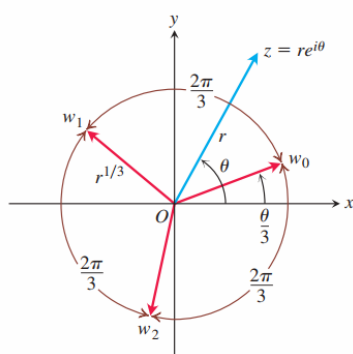


FIGURE 18.6 The three cube roots of $z = re^{i\theta}$



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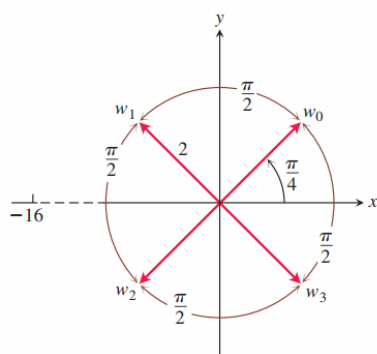


FIGURE 18.7 The four fourth roots of -16

EXAMPLE 7 Find the four fourth roots of -16 .

Solution As our first step, we plot the number -16 in the complex plane (Figure 18.7) and determine its polar representation $re^{i\theta}$. Here, $z = -16$, $r = +16$, and $\theta = \pi$. One of the fourth roots of $16e^{i\pi}$ is $2e^{i\pi/4}$. We obtain others by successive additions of $2\pi/4 = \pi/2$ to the argument of this first one. Hence,

$$\sqrt[4]{16e^{i\pi}} = 2e^{ik\pi/4}, \quad k = 1, 3, 5, 7, \tag{12}$$

and the four roots are

$$\begin{aligned} w_0 &= 2\left[\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right] = \sqrt{2}(1 + i), \\ w_1 &= 2\left[\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right] = \sqrt{2}(-1 + i), \\ w_2 &= 2\left[\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right] = \sqrt{2}(-1 - i), \\ w_3 &= 2\left[\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}\right] = \sqrt{2}(1 - i). \end{aligned}$$

The Fundamental Theorem of Algebra

One might say that the invention of $\sqrt{2}$ is all well and good and leads to a number system that is richer than the rational number system alone, and the complex number system is richer still, but where will this process end? Will we need to develop still more systems so as to obtain $\sqrt[4]{-1}$ (a solution of $x^4 = -1$), $\sqrt[5]{-1}$ (a solution of $x^5 = -1$), and so on? It turns out this is not necessary. These numbers are already expressible in the complex number system as $a + ib$ for appropriate a and b . In fact, the Fundamental Theorem of Algebra says that with the introduction of the complex numbers, we now have enough numbers to factor every polynomial into a product of linear factors. We have enough numbers to solve every possible polynomial equation.

The Fundamental Theorem of Algebra

Every polynomial equation of the form

$$a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0 = 0,$$

in which the coefficients $a_0, a_1, \dots, a_{n-1}, a_n$ are any complex numbers, whose degree n is greater than or equal to one, and whose leading coefficient a_n is not zero, has exactly n roots in the complex number system, provided each multiple root of multiplicity m is counted as m roots.

A proof of this theorem can be found in most texts on the theory of functions of a complex variable.

EXERCISES 18.1

Operations with Complex Numbers

- Express the following products of complex numbers in the form $a + ib$ where a and b are real numbers.
 - $(2 + 3i)(4 - 2i)$
 - $(2 - i)(-2 - 3i)$
 - $(-1 - 2i)(2 + i)$
- Solve the following equations for the real numbers x and y .
 - $(3 + 4i)^2 - 2(x - iy) = x + iy$
 - $\left(\frac{1+i}{1-i}\right)^2 + \frac{1}{x+iy} = 1 + i$
 - $(3 - 2i)(x + iy) = 2(x - 2iy) + 2i - 1$





18.2 Functions of a Complex Variable

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Graphing and Geometry

3. How may the following complex numbers be obtained from $z = x + iy$ geometrically? Sketch.
a. $\bar{z}$ b. $-\bar{z}$ c. $-z$ d. $1/z$
4. Show that the distance between the two points z_1 and z_2 in the complex plane is $|z_1 - z_2|$.

In Exercises 5–10, graph the points $z = x + iy$ that satisfy the given conditions.

5. a. $|z| = 2$ b. $|z| < 2$ c. $|z| > 2$
6. $|z - 1| = 2$ 7. $|z + 1| = 1$
8. $|z + 1| = |z - 1|$ 9. $|z + i| = |z - i|$
10. $|z + 1| \geq |z - 1|$

Express the complex numbers in Exercises 11–14 in the form $re^{i\theta}$, with $r \geq 0$ and $-\pi < \theta \leq \pi$, and for each of them draw a diagram in the complex plane.

11. $(1 + \sqrt{-3})^2$ 12. $\frac{1+i}{1-i}$
13. $\frac{1+\sqrt{-3}}{1-\sqrt{-3}}$ 14. $(2+3i)(1-2i)$

Powers and Roots

Use De Moivre's Theorem to express the trigonometric functions in Exercises 15 and 16 in terms of $\cos \theta$ and $\sin \theta$.

15. $\cos 4\theta$ 16. $\sin 4\theta$
17. Find the three cube roots of 1.
18. Find the two square roots of i .
19. Find the three cube roots of $-8i$.
20. Find the six sixth roots of 64.
21. Find the four solutions of the equation $z^4 - 2z^2 + 4 = 0$.
22. Find the six solutions of the equation $z^6 + 2z^3 + 2 = 0$.

23. Find all solutions of the equation $x^4 + 4x^2 + 16 = 0$.
24. Solve the equation $x^4 + 1 = 0$.

Theory and Examples

25. **Complex numbers and vectors in the plane** Show with a diagram in the complex plane that the law for adding complex numbers is the same as the parallelogram law for adding vectors.
26. **Complex arithmetic with conjugates** Show that the conjugate of the sum (product, or quotient) of two complex numbers, z_1 and z_2 , is the same as the sum (product, or quotient) of their conjugates.
27. **Complex roots of polynomials with real coefficients come in complex-conjugate pairs**
a. Extend the results of Exercise 26 to show that $f(\bar{z}) = \overline{f(z)}$ when

$$f(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

is a polynomial with real coefficients $a_0, \dots, a_n$.

- b. If z is a root of the equation $f(z) = 0$, where $f(z)$ is a polynomial with real coefficients as in part (a), show that the conjugate $\bar{z}$ is also a root of the equation. (Hint: Let $f(z) = u + iv = 0$; then both u and v are zero. Use the fact that $f(\bar{z}) = \overline{f(z)} = u - iv$.)
28. **Absolute value of a conjugate** Show that $|\bar{z}| = |z|$.
29. **When $z = \bar{z}$** If z and $\bar{z}$ are equal, what can you say about the location of the point z in the complex plane?
30. **Real and imaginary parts** Show that the following relations hold for any complex numbers z, z_1 , and z_2 .
a. $z + \bar{z} = 2\operatorname{Re}(z)$
b. $z - \bar{z} = 2i\operatorname{Im}(z)$
c. $2\operatorname{Re}(z) \leq |z|$
d. $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2\operatorname{Re}(z_1 \bar{z}_2)$
e. $|z_1 + z_2| \leq |z_1| + |z_2|$

18.2 Functions of a Complex Variable

Complex Domains

The domains of the real variable functions discussed in Chapter 1 consisted of subsets of the real line, such as $[0, 1]$ or $(0, \infty)$. The domain of a complex function is a subset of the complex numbers, which we can visualize as a region in the complex plane.

EXAMPLE 1 Examples of complex domains

- a. The set of complex numbers z for which $|z| \leq 1$
The corresponding region in the complex plane is the set of points lying inside or on a circle of radius 1 centered at the origin O , shown in Figure 18.8(a).
- b. The set of complex numbers with $|z - z_0| \leq 1$
This corresponds to the region inside or on a circle of radius 1 centered at a complex number $z_0 = a_0 + b_0 i$, as in Figure 18.8b.
- c. A complex region can have a more general shape. Three more examples of complex domains are shown in Figure 18.8c.

